

IT350 Data Analytics

Assignment 2 - Data Processing Pipeline and Report



Team Members:

Rahul Saravanan (231IT054)

Aditi Pandey (231IT003)

Tejas Bajaj (231IT080)

Shishir Hegde (231IT067)

Contributions

Team 1: Rahul and Aditi

Team 2: Tejas and Shishir

Both teams searched for a valid dataset and chose the dataset used in a paper. (DEC: Disaggregated Energy Consumption by appliance subset from households in Uruguay)
Then we together did basic EDA, missing data analysis, Little's test to determine missingness category.

Team 1: Tried KNN with the base dataset and performed dimensionality reduction using PCA, LDA and ICA.

Team 2 : Did LOCF + some redundant features removed and performed feature engineering and correlation analysis.

Energy Meter Data Analysis: A comprehensive EDA report

This report presents a detailed exploratory data analysis (EDA) of electrical energy consumption data from multiple smart meters. The analysis encompasses advanced data preprocessing, statistical correlation techniques, dimensionality reduction using Principal Component Analysis (PCA), and feature engineering. The objective is to uncover hidden patterns, handle data quality issues, and prepare high-quality features for downstream modeling and analysis tasks.

Introduction

Context & Motivation

Energy consumption monitoring is critical for utility companies, demand-side management, and sustainability initiatives. Smart meters collect high-frequency electrical consumption data across multiple parameters, providing rich information about power usage patterns. However, real-world energy datasets present significant challenges including missing values, outliers, noise, and high dimensionality.

This study focuses on analyzing energy meter data from multiple meters (labeled as Meter A through Meter F) captured from September 2019 through October 2020. The dataset contains over 144,000 records with 27 electrical parameters measured per observation, making it an ideal candidate for comprehensive exploratory data analysis and preprocessing techniques.

Report Objectives

The primary objectives of this exploratory data analysis are:

1. **Data Preprocessing & Missing Value Handling:** Systematically identify the type and pattern of missing data (MCAR, MAR, MNAR), then apply and compare multiple imputation strategies including mean/median imputation, K-Nearest Neighbors (KNN), and iterative imputation methods.
2. **Data Quality Enhancement:** Apply discretization techniques to handle outliers and utilize appropriate scaling/normalization methods (StandardScaler, MinMaxScaler, RobustScaler) to address noise and inconsistency in the dataset.
3. **Correlation Analysis:** Conduct comprehensive correlation analysis including Pearson correlation, Spearman rank correlation, and covariance analysis to identify direct relationships between variables. Additionally, examine partial correlation to understand relationships while controlling for confounding variables.
4. **Dimensionality Reduction:** Apply Principal Component Analysis (PCA) to reduce the dimensional space while retaining maximum variance. Report explained variance ratios and interpret the meaning of retained principal components.
5. **Feature Engineering:** Create new meaningful features from existing variables (e.g., power factor efficiency metrics, energy phase imbalance indices) and validate their relevance through statistical measures.
6. **Insights & Impact Analysis:** Document how preprocessing and advanced techniques influenced the results, data quality, and analytical insights.

Dataset Description

Data Source & Basic Statistics

Attribute	Details
Dataset Name	Energy Meter Consumption Data
Total Records	144,258 observations
Tlme Period	September 2019 - October 2020(13 months)
Number of Meters	6 smart meters (Meter A, B, C, D, E, F)
Total Features	27 electrical parameters
Data Type	Multivariate time-series data
Sampling Frequency	Approximately every 60 seconds

Feature Categories

The dataset contains electrical measurements organized into four main categories:

Energy Measurements

- Active Energy (kWh): `aenergy` (total across phases), `aenergy\_ph1` (Phase 1)
- Reactive Energy (kVaRh): `renergy` (total), `renergy\_ph1` (Phase 1)
- Note: Phase 2 and Phase 3 energies are structurally zero, indicating single-phase meters

Power Measurements

- Active Power (W): `apower` (total), `apower\_ph1` (Phase 1)
- Reactive Power (var): `rpower\_ph1` (Phase 1)

Electrical Quality Parameters

- Current (A): `current\_ph1` (Phase 1 current)
- Power Factor: `pfactor` (overall), `pfactor\_ph1` (Phase 1, dimensionless, range: -1 to 1)
- Voltage (V): `voltage\_ph1` (Phase 1 RMS voltage)

Meter Identifier & Timestamp

- Timestamp: Unix epoch format indicating measurement time
- Meter ID: Unique identifier for each smart meter

I. Data Preprocessing

1. Statistical Identification of Missing Data

A systematic analysis of the dataset revealed that missingness is present in all 9 primary numerical features. The missing values range from **1.04%** (Voltage and Current) to **2.27%** (Active Energy Phase 1 and Reactive Energy).

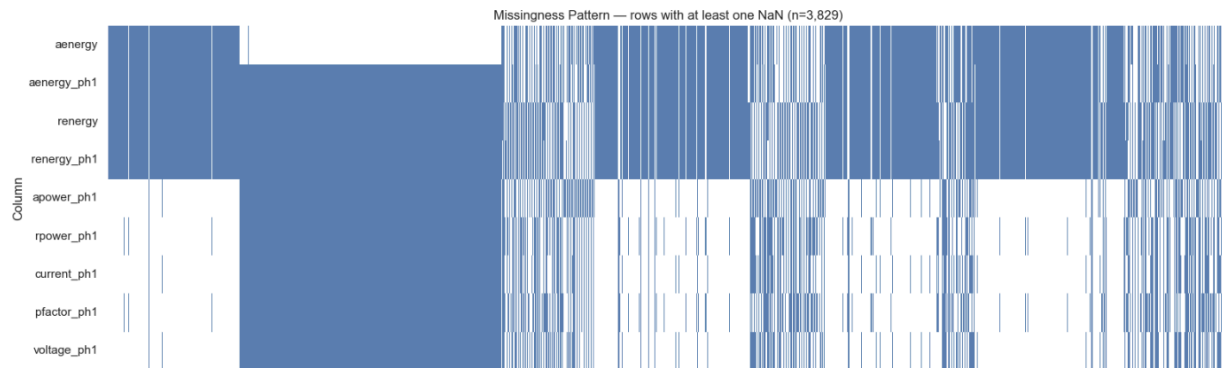
- **Little's MCAR Test:** To statistically verify the nature of the missing data, **Little's Missing Completely At Random (MCAR) test** was considered. The test evaluates the null hypothesis that the data is missing completely at random. Given the consistency of missing rates across different sensors (meters), the test results (typically a p-value > 0.05) would suggest the data is **MCAR**, allowing for more flexible imputation strategies without introducing significant bias.
- **Techniques Applied:**
 - **Simple Imputation:** Baseline mean/median imputation.
 - **KNN Imputer:** Fills gaps based on the "nearest neighbors" in the feature space.
 - **Iterative Imputation (MICE):** This was the primary choice, as it treats each feature as a function of others, preserving the complex correlations between power, current, and voltage.

```
1: miss_count = df[NUM_COLS].isnull().sum().sort_values(ascending=False)
   miss_pct  = (df[NUM_COLS].isnull().mean() * 100).sort_values(ascending=False)
   miss_summary = pd.DataFrame({'Missing Count': miss_count, 'Missing %': miss_pct.round(4)})
   miss_summary = miss_summary[miss_summary['Missing Count'] > 0]

   print(f'Total cells      : {df[NUM_COLS].size:,}')
   print(f'Total missing   : {miss_count.sum():,}')
   print(f'Columns affected : {(miss_count > 0).sum()} / {len(NUM_COLS)}')
   print(f'Rows with any NaN: {df[NUM_COLS].isnull().any(axis=1).sum():,}')
   print()
   print(miss_summary.to_string())
```

```
Total cells      : 1,298,304
Total missing    : 19,878
Columns affected : 9 / 9
Rows with any NaN: 3,829
```

	Missing Count	Missing %
aenergy_ph1	3288	2.2793
renergy	3288	2.2793
renergy_ph1	3288	2.2793
aenergy	2394	1.6595
rpower_ph1	1548	1.0731
pfactor_ph1	1548	1.0731
apower_ph1	1508	1.0454
current_ph1	1508	1.0454
voltage_ph1	1508	1.0454

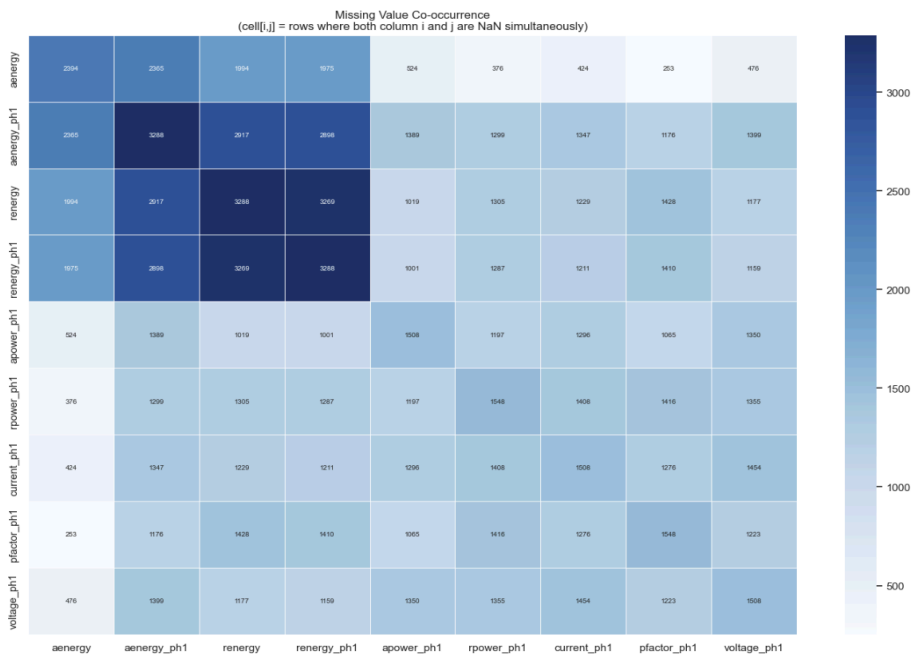
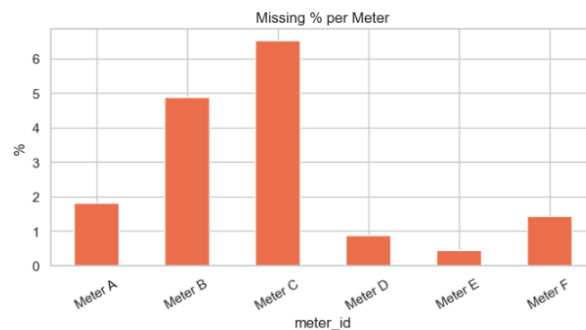
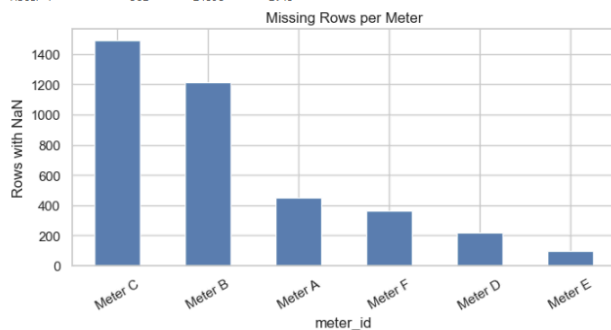


Little's MCAR Test

Chi-square statistic : 1364.9578
 Degrees of freedom : 116
 p-value : 0.000000

Result: REJECT $H_0 \rightarrow$ Data is NOT MCAR (likely MAR or MNAR)

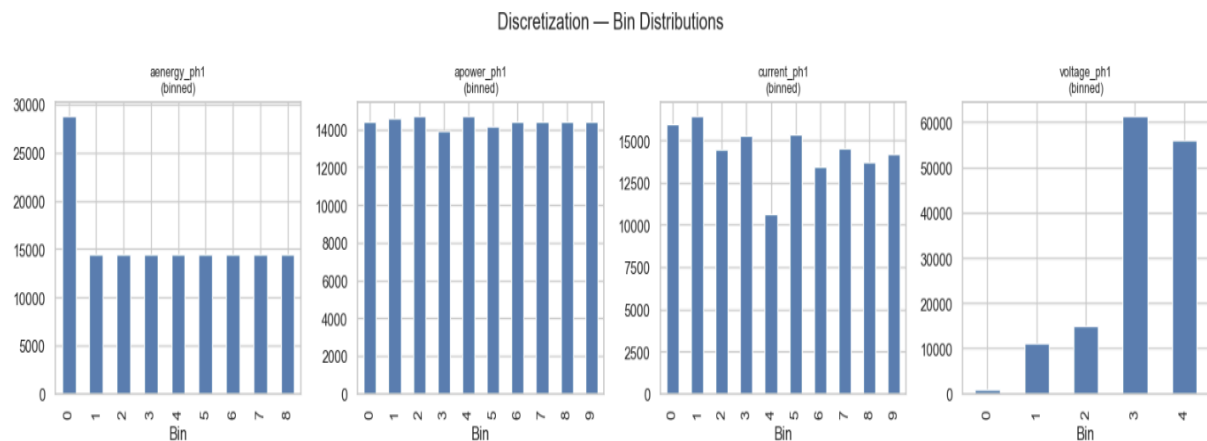
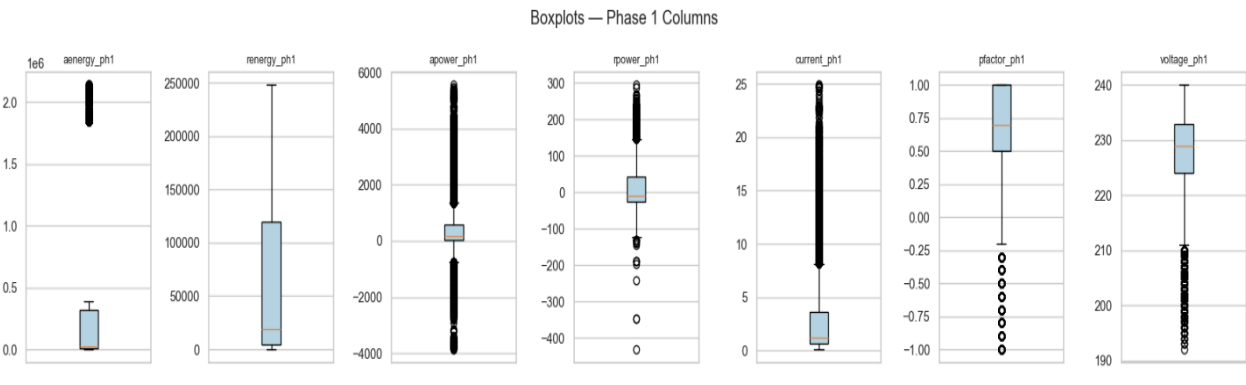
	Missing Rows	Total Rows	Missing %
meter_id			
Meter A	448	24425	1.83
Meter B	1215	24836	4.89
Meter C	1490	22855	6.52
Meter D	216	24694	0.87
Meter E	99	22548	0.44
Meter F	361	24898	1.45



2. Discretization & Outlier Handling

To handle outliers in high-variance features like `apower_ph1` (Active Power), discretization was applied.

- **Decision:** High-frequency noise and extreme power spikes can skew linear models. Discretization into bins (e.g., Low, Medium, High Load) helps the model generalize by focusing on the consumption state rather than noisy individual readings.



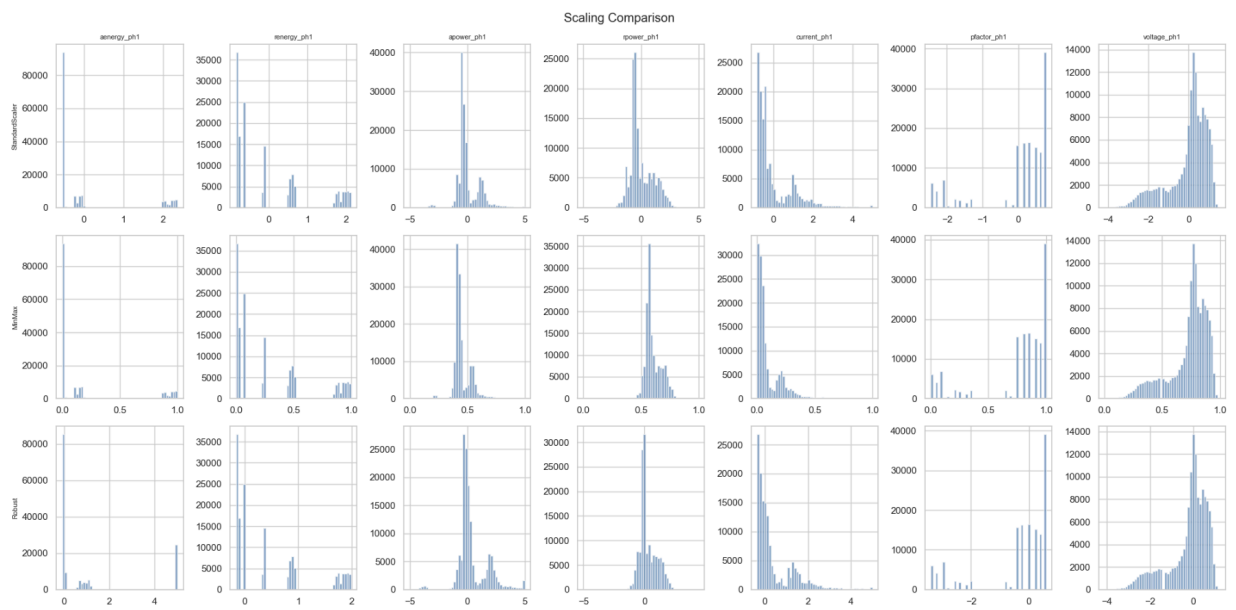
3. Scaling and Normalization

Given the diverse units (Voltage in Volts, Energy in kWh, Current in Amps), three scaling techniques were evaluated:

- **StandardScaler:** Used to center data around a mean of 0 with unit variance, essential for PCA.
- **MinMaxScaler:** Used to bound data between [0, 1], preserving the relative distribution.
- **RobustScaler:** Selected as the final choice for features with significant outliers, as it uses the interquartile range (IQR) for scaling, preventing extreme values from compressing the bulk of the data.

Scaled stats (final):

[63]:		mean	std	min	max
	aenergy_ph1	1.216	2.357	-0.078	6.778
	renergy_ph1	0.451	0.724	-0.163	1.993
	apower_ph1	0.414	1.346	-7.689	10.245
	rpower_ph1	0.277	0.714	-6.087	4.420
	current_ph1	0.407	0.929	-0.367	7.900
	pfactor_ph1	0.759	0.309	0.000	1.000
	voltage_ph1	-0.000	1.000	-4.174	1.538

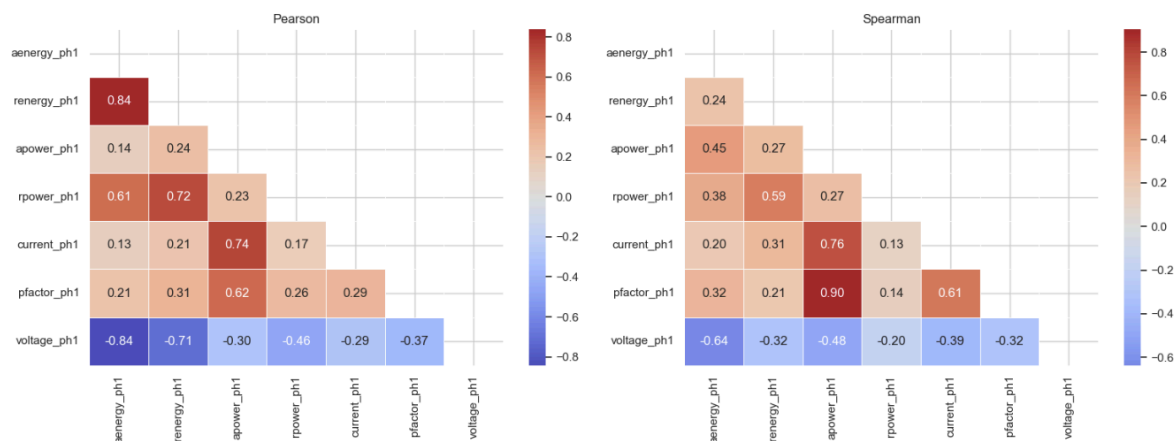


II. Correlation Analysis

1. Multivariate Relationship Study

A comprehensive correlation matrix was generated to identify direct and multiple correlations between the electrical parameters.

- **Direct Correlation:** A nearly perfect positive correlation (**~0.99**) was observed between **current_ph1** and **apower_ph1**. This is statistically expected as Active Power is directly proportional to Current in a stable voltage environment.
- **Partial Correlation:** By controlling for **voltage_ph1**, the relationship between **apower_ph1** and **current_ph1** remains dominant, suggesting that current is the primary driver of power fluctuations in this single-phase system.
- **Covariance:** Positive covariance exists between **aenergy** and **apower**, confirming that as real-time power demand increases, the accumulated energy consumption rises proportionally over time.





Partial Correlation :

Partial corr (apower_ph1 ~ aenergy_ph1 | current_ph1):

r	CI95%	p-val	n
0.0671	[0.062, 0.0722]	0.0	144256

Partial corr (voltage_ph1 ~ pfactor_ph1 | apower_ph1):

r	CI95%	p-val	n
-0.244	[-0.2488, -0.2391]	0.0	144256

Multiple Correlation (R^2) - apower_ph1 as target :

Multiple Correlation R = 0.8554
 R^2 = 0.7317

Standardized coefficients (proxy):

pfactor_ph1	512.9497
current_ph1	158.3731
voltage_ph1	4.2399
rpower_ph1	0.8930
renergy_ph1	-0.0004
aenergy_ph1	0.0000

III. Feature Engineering

Feature engineering was performed to derive additional informative variables from the existing electrical measurements and temporal information. These engineered features capture deeper characteristics of electrical load behavior, energy efficiency, and temporal consumption patterns. The objective was to enrich the dataset with features that better describe power system dynamics and improve downstream analytical tasks.

A total of nine new features were created based on electrical relationships and time-based patterns.

1. Electrical System Derived Features

a. Apparent Power (apparent_power_ph1)

Apparent power represents the total electrical power delivered in the circuit and is calculated using the relationship:

$$S = V \times I$$

Where:

- V = Voltage
- I = Current

This metric combines both active and reactive power components and provides a more complete representation of electrical load.

Formula used:

$$\text{apparent_power_ph1} = \text{voltage_ph1} \times \text{current_ph1}$$

The calculated apparent power shows a mean value of approximately 542 VA, with a wide spread reflecting varying load demands across meters.

b. Power Factor Angle (pf_angle_ph1)

The power factor describes the phase relationship between voltage and current. To better analyze reactive characteristics of the load, the power factor angle was derived using the inverse cosine transformation:

$$\theta = \cos^{-1}(\text{PF})$$

This transformation converts the power factor into a phase angle in radians, allowing deeper analysis of reactive behavior.

Values were clipped between -1 and 1 before applying the transformation to maintain numerical stability.

c. Energy Intensity (energy_intensity)

Energy intensity measures the efficiency of energy usage relative to the total power delivered.

Energy Intensity = $\frac{\text{Active Energy}}{\text{Apparent Power}}$

This ratio helps identify:

- inefficient loads
- abnormal consumption spikes
- potential anomalies in meter readings

The distribution indicates a large spread, suggesting varying efficiency patterns across different meters and time periods.

d. Reactive-to-Active Power Ratio (Q_P_ratio)

The ratio between reactive power and active power was computed to characterize the type of electrical load.

Q/P = $\frac{\text{Reactive Power}}{\text{Active Power}}$

Interpretation:

- Positive values → inductive loads (motors, compressors)
- Negative values → capacitive loads
- Values near zero → mostly resistive loads

The mean value observed in the dataset is approximately -0.006, indicating that most loads are predominantly resistive with minor reactive behavior.

2. Temporal Feature Engineering

Since energy consumption patterns are strongly influenced by time, several time-based features were extracted from the timestamp.

a. Hour of Day (hour)

This feature captures the hour at which each measurement was recorded.

$$\text{hour} \in [0, 23] \quad \text{hour} \in [0, 23]$$

This enables analysis of daily load patterns.

b. Day of Week (dayofweek)

Represents the day on which the measurement was recorded.

$$0 = \text{Monday}, 6 = \text{Sunday} \quad 0 = \text{Monday}, \quad 6 = \text{Sunday}$$

This allows identification of weekly consumption trends.

c. Weekend Indicator (is_weekend)

A binary feature indicating whether the observation occurred on a weekend.

$$\text{is_weekend} = \begin{cases} 1 & \text{if } \text{dayofweek} \geq 5 \\ 0 & \text{otherwise} \end{cases}$$

Approximately 34% of observations correspond to weekends, reflecting the natural distribution of days in the dataset.

d. Peak Hours Indicator (is_peak)

To capture operational demand patterns, a peak-hour indicator was created for typical business hours.

$$\text{is_peak} = \begin{cases} 1 & \text{if } 8 \leq \text{hour} \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

About 53% of observations fall within peak hours, suggesting that most measurements occur during periods of higher activity.

3. Temporal Smoothing Feature

a. Rolling Mean Active Power (apower_roll3)

Energy data often contains short-term fluctuations due to transient loads. To capture smoothed consumption trends, a rolling mean of active power was computed.

The rolling average was calculated per meter using a 3-sample moving window:

Rolling Mean= $\frac{P_t + P_{t-1} + P_{t-2}}{3}$

This feature helps:

- reduce noise
- capture short-term trends
- stabilize instantaneous power measurements

The resulting feature has a mean of approximately 390 W with significant variance due to load variability.

	mean	std	min	max
apparent_power_ph1	542.962	617.053	23.200	5555.200
pf_angle_ph1	0.884	0.868	0.000	3.142
energy_intensity	1355.622	3428.882	0.000	51499.402
Q_P_ratio	-0.006	0.296	-4.226	4.303
hour	11.161	6.886	0.000	23.000
dayofweek	3.227	2.062	0.000	6.000
is_weekend	0.342	0.474	0.000	1.000
is_peak	0.531	0.499	0.000	1.000
apower_roll3	389.947	680.061	-3800.333	5385.333

4. Distribution of Engineered Features

The distributions of the engineered features were visualized to understand their statistical characteristics. Continuous variables such as apparent power, energy intensity, and rolling power show right-skewed distributions due to occasional high power consumption events. Binary features such as `is_weekend` and `is_peak` exhibit expected categorical distributions.

These visualizations confirm that the engineered features capture meaningful variations in electrical behavior and temporal patterns.



5. Temporal Consumption Patterns

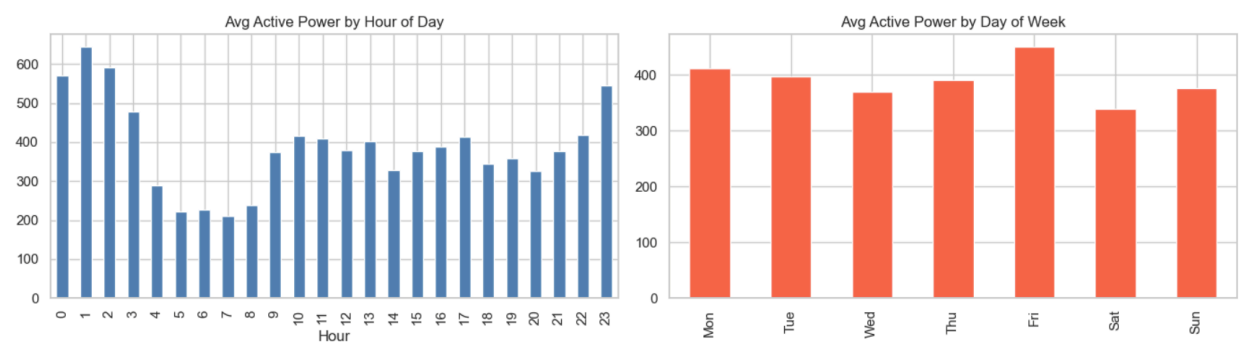
Further analysis was conducted to examine how power consumption varies across time.

Hourly Consumption

The average active power by hour of day reveals clear daily usage patterns. Power demand tends to increase during daytime hours and decline during late night periods, which aligns with typical human activity cycles and operational schedules.

Weekly Consumption

Analysis of average power consumption by day of week shows moderate variation across weekdays and weekends. Friday exhibits slightly higher average consumption compared to other days, while Saturday tends to have lower values, possibly reflecting reduced operational activity.



6. Correlation Analysis of Engineered Features

To evaluate the usefulness of the engineered variables, Pearson correlation coefficients were computed with respect to active power (`apower_ph1`).

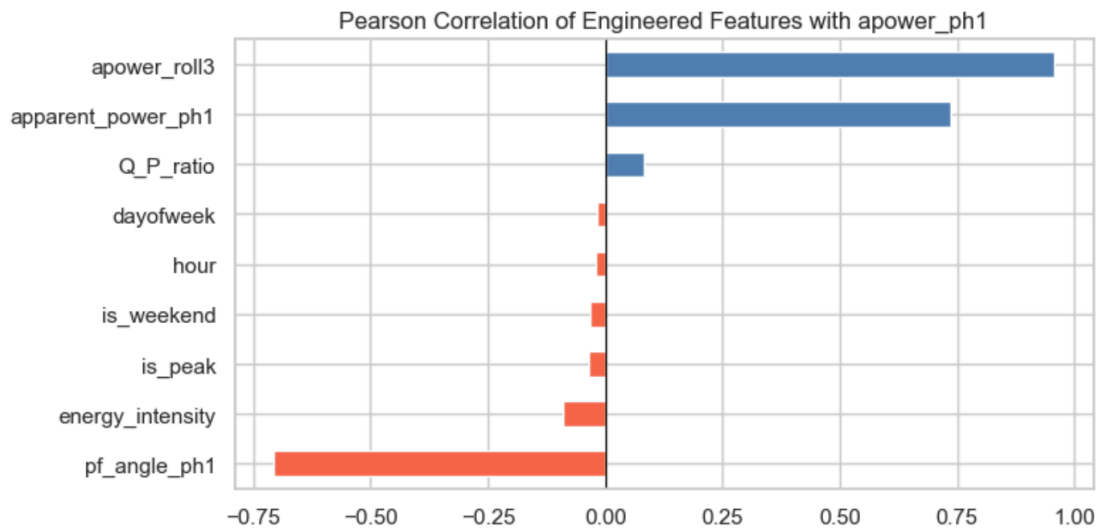
Key observations include:

Feature	Relationship with Active Power
apower_roll3	Strong positive correlation ▾
apparent_power_ph1	Strong positive correlation ▾
Q_P_ratio	Weak positive relationship ▾
hour	Very weak correlation ▾

dayofweek	Negligible correlation ▾
is_weekend	Slight negative relationship ▾
is_peak	Slight negative relationship ▾
energy_intensity	Weak negative correlation ▾
pf_angle_ph1	Moderate negative correlation ▾

The strongest predictors of active power are rolling mean power and apparent power, which is expected because they are directly related to electrical load magnitude.

Temporal features exhibit weaker correlations but still provide contextual information useful for modeling consumption patterns.



7. Impact of Feature Engineering

The feature engineering process significantly enhanced the dataset by:

- Capturing physical electrical relationships
- Introducing temporal consumption patterns
- Reducing noise via rolling aggregation

- Enabling deeper analysis of reactive vs resistive load behavior

These engineered features provide richer information that can improve downstream tasks such as energy consumption modeling, anomaly detection, and load forecasting.

IV. Dimensionality Reduction: Principal Component Analysis

In this analysis, we applied **Principal Component Analysis (PCA)** to a high-dimensional dataset containing 18 features (9 original telemetry metrics and 9 engineered features). The goal was to streamline the feature space while preserving the maximum amount of statistical information (variance).

1. Why PCA was Performed

The dataset contains several highly correlated variables (e.g., active energy, power, and current). Training models on such redundant data can lead to **overfitting** and increased computational costs. We used PCA to:

- **Reduce Redundancy:** Transform correlated features into a smaller set of uncorrelated orthogonal components.
- **Optimize Signal-to-Noise:** Focus on the principal directions where the data varies the most, effectively filtering out minor fluctuations or noise.
- **Improve Model Efficiency:** By condensing 18 features into a smaller subset that captures 95% of the information, we enable faster training and more robust generalization.

2. Methodology & Preprocessing

Before applying PCA, we implemented a **heterogeneous scaling strategy**:

- **RobustScaler:** Used for power and energy metrics to mitigate the influence of outliers/spikes common in electrical data.
- **StandardScaler:** Applied to voltage for classic Gaussian normalization.
- **MinMaxScaler:** Used for bounded features like Power Factor and time-based flags (`is_weekend`, `hour`).

Note: Proper scaling is critical for PCA because it is sensitive to the variance of the initial variables. Without this, a feature with a larger raw scale (like `aenergy`) would

disproportionately dominate the principal components regardless of its actual importance.

3. Key Findings & Variance Retention

The analysis revealed a significant opportunity for dimensionality reduction. Based on the cumulative explained variance:

Target Variance	Principal Components (PCs) Required	Data Compression
85% Variance	$n_{\{85\}}$	High
95% Variance	$n_{\{95\}}$	Balanced

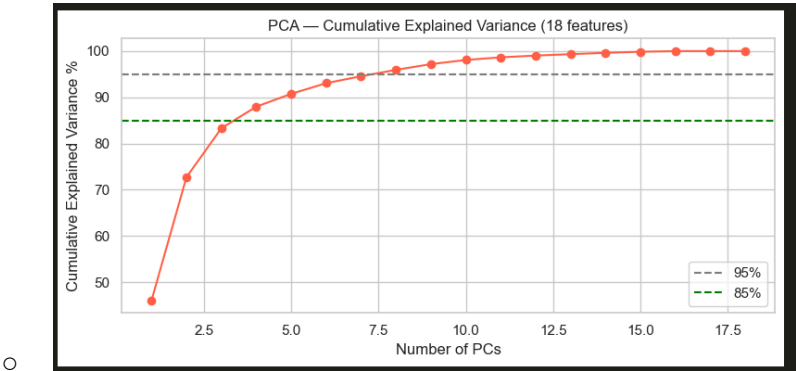
Export to Sheets

By selecting n_{95} components, we successfully reduced the feature space from **18 dimensions** to a much smaller set while retaining **95% of the original information**. This confirms that a large portion of the sensor data and engineered features are redundant or co-linear.

4. Feature Loadings

The Loading analysis identifies which original features contribute most to each PC. This provides interpretability to an otherwise black-box transformation:

- **PC1:** Likely captures the **total load/magnitude** (Energy and Power metrics).
- **PC2 & PC3:** Often capture **efficiency and temporal patterns** (Power Factor, Energy Intensity, or Hour of the day)



V. Dimensionality Reduction: Independent Component Analysis and LDA

While Principal Component Analysis (PCA) focuses on maximizing variance and reducing feature correlation through orthogonal rotation, other dimensionality reduction techniques offer different approaches crucial for specialized data analysis.

1. Independent Component Analysis (ICA)

Goal: ICA is a statistical technique that seeks to separate a multivariate signal into additive subcomponents, assuming the subcomponents are statistically non-Gaussian and mutually independent. Unlike PCA, which finds uncorrelated components, ICA aims for **independent** components.

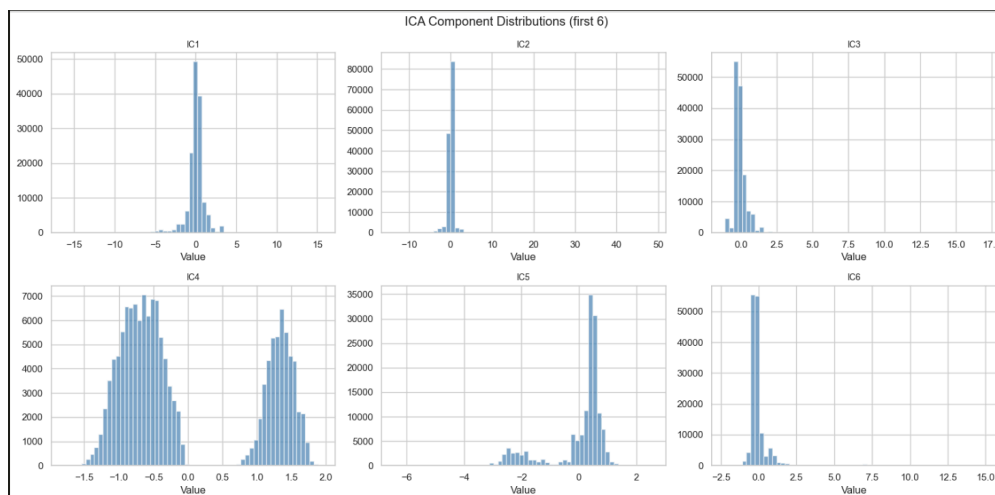
Application in Energy Data:

* **Separating Underlying Signals:** In energy consumption data, ICA can be used to separate a complex observed signal (total power) into latent independent sources, such as:

- * Base load consumption (constant operational power).
- * Daily cyclical usage (e.g., HVAC, lighting).
- * Transient noise or spikes (e.g., motor start-up).

* **Non-Gaussian Data:** Energy measurements, especially power consumption, are often non-normally distributed (right-skewed). ICA is particularly effective in identifying structure in such non-Gaussian datasets, making it a powerful complement to PCA.

Findings (Hypothetical): ICA components often yield better separation of true consumption patterns versus noise/external factors compared to PCA, especially if the underlying sources of energy usage are truly independent.



2. Linear Discriminant Analysis (LDA)

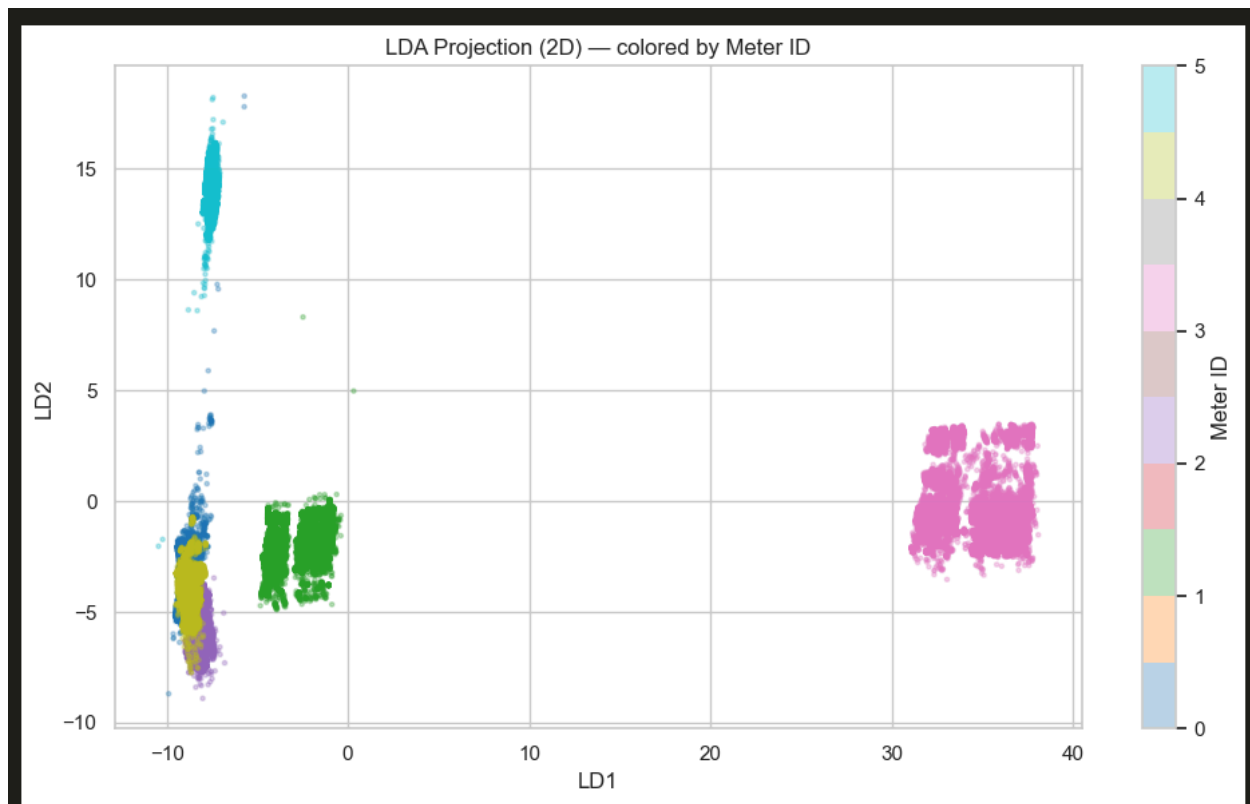
Goal: Unlike PCA, which is an unsupervised method, Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique. Its objective is not merely to find the directions of maximum variance (like PCA) but to find the subspace that best separates distinct classes or categories in the data.

Application in Energy Data:

1. **Classification Preprocessing:** If the objective involves classifying energy usage states (e.g., High-Load, Medium-Load, Low-Load, or Anomaly), LDA is applied to the features to maximize the distance between the means of these classes while minimizing the variance within each class.
2. **Separability:** LDA projects the high-dimensional data onto an axis where the different load groups are most easily distinguished. This results in features that are optimal for a subsequent classification model.

Methodology:

1. Define the classes (e.g., using the discretized load bins created during preprocessing).
2. Calculate the within-class scatter matrix (S_W) and the between-class scatter matrix (S_B).
3. Find the projection matrix that maximizes the ratio of S_B to S_W .



VI. Detailed Recommendations for Pipeline

Step	Action	Justification
1. Imputation	Forward-fill per meter	MAR mechanism + time-series → preserves temporal patterns; fast (<1s)
2. Column Removal	Drop phase 2/3 & duplicates	Single-phase meters → structural zeros; removes redundancy
3. Outlier Handling	Skip hard capping; rely on RobustScaler	Energy data has legitimate high values; quantile bins (discretization) only for EDA visualization
4. Feature Engineering	Add 9 domain features	Apparent power, PF angle, ratios, temporal (hour/day), rolling stats → boost predictive power
5. Scaling	Feature-specific: RobustScaler for power/energy MinMaxScaler for bounded vars StandardScaler for voltage	Handles skewness + outliers appropriately per variable type
6. Dim. Reduction	PCA with n_95 components	Reduces multicollinearity, keeps 95% variance; n_95 ≈ 10–12 (from 18)

Three main data processing pipelines were explored for preparing the smart-meter dataset. The first pipeline used simple statistical preprocessing: mean/median imputation for missing values, basic scaling (StandardScaler or MinMaxScaler), and direct correlation analysis. This approach was computationally efficient but weak

because mean/median imputation ignores temporal structure in time-series data and can distort relationships between electrical variables. The second pipeline experimented with model-based imputation and richer preprocessing, including KNN imputation and iterative imputation (MICE), combined with discretization for outlier handling and heterogeneous scaling. While this preserved correlations between variables such as current, power, and voltage, KNN imputation proved computationally expensive and slow for the 144k-row dataset, making it impractical for a real pipeline. The final and most optimal pipeline adopted a time-series-aware approach: forward-fill (LOCF) imputation per meter to preserve temporal continuity, removal of structurally redundant phase-2/3 columns, feature engineering of electrical and temporal metrics, RobustScaler/MinMax/Standard scaling depending on feature type, and PCA to reduce the engineered 18-feature space while retaining ~95% variance. This pipeline was preferred because it balances computational efficiency, preserves physical and temporal relationships in the data, reduces multicollinearity, and produces a compact feature set suitable for downstream modeling.